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Review article

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Advances in the use of virtual reality to treat mental health conditions

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Abstract

Virtual reality (VR) has emerged as a promising tool in the field of mental health. Central to this technology are immersive environments, which enable exposure to highly controlled virtual experiences that feel real. In this Review, we elaborate on the active elements of immersive experiences and how VR-based treatments work. We provide an overview of developments in the use of VR to treat mental health conditions (anxiety, psychotic symptoms, post-traumatic stress, eating disorders, depression and stress management) with a focus on the core mechanisms that drive effective interventions. Artificial intelligence, biofeedback and gamification are emerging areas of development, and we discuss how they might enhance the accessibility, engagement and effectiveness of psychological treatments. Conducting rigorous studies with user-centred designs in diverse populations is a key research priority. As the use of VR in mental health continues to evolve, addressing ethical and implementation considerations is critical for ensuring ongoing treatment improvements.

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Introduction

Psychological interventions seek to help individuals to manage symptoms and improve functioning in everyday life. However, interventions are typically delivered in clinical settings in the absence of relevant triggers, away from the real-world contexts where mental health challenges play out. This gap between therapy and real-world experiences can limit ecological validity (the degree to which research methods and findings reflect behaviours and phenomena in everyday settings) and hamper treatment effectiveness. Virtual reality (VR) overcomes these challenges by enabling the recreation of real-world environments with a high degree of precision and control[1]. In virtual environments, individuals can learn and adapt to relevant situations and scenarios in real time, progressively and safely under the supervision of a therapist[2].

At its core, VR offers possibilities to simulate reality that simply did not exist with the traditional methods previously available[3]. Through a head-mounted display that projects 3D images onto two screens (one for each eye), VR users are immersed in a computer-generated virtual environment that can closely simulate the real world (Fig. 1). Audio and hand-tracking sensors enable interaction with objects and virtual avatars, which might be computer generated, or with other VR users. The immersive environment and real-time tracking sensors represent transformative opportunities for human interaction and experience, including for mental health treatment[4]. Furthermore, as VR technologies grow increasingly accessible in modern society, harnessing these tools as a vehicle for remote treatment provides an avenue for enhancing the accessibility and scalability of mental health solutions. Improving access to mental health treatment is a critical and urgent unmet need in the face of rising rates of mental health conditions globally[5].

Research evidence is mounting in favour of the potential clinical utility of VR – along with online therapies, smartphone apps and wearables – for clinical research, assessment and intervention[6–10], particularly when combined with human support[11]. Indeed, VR has emerged as a highly promising technology that can improve the quality and reach of available treatments[7,12]. Through persuasive systems design and gamification, VR and other digital mental health technologies can also address barriers to help-seeking and poor engagement[7], particularly amongst youth populations who lead increasingly digital lives and are interested in using technologies to support their mental health[1]. There has been substantial growth in VR research since the mid 2010s driven by promising research findings and associated industry attention to the potential commercial viability of these platforms[13].

In this Review, we provide up-to-date evidence on the application of VR for the treatment of mental health conditions, highlight advances and challenges in the field, and identify key priorities for future research. To that end, we incorporate evidence from comprehensive systematic reviews and meta-analyses of immersive VR-based treatments[9,14–17]. First, we summarize the core elements and mechanisms of immersive VR-based treatments, with particular considerations around motivation and sense of presence. Subsequently, we describe the evidence for the effectiveness of VR-based treatments across mental health conditions, prioritizing conditions in which the evidence is well established (anxiety, psychosis, post-traumatic stress disorder (PTSD), eating disorders, depression and stress management). We identify and describe the major emerging areas of innovation and development in VR-based treatments, including artificial intelligence, biofeedback and gamification. Finally, we summarize the evidence base and discuss priorities for future research.

Our Review does not cover evidence about VR-based assessment, which is being researched and reviewed elsewhere[6]. VR has been investigated for the treatment of developmental disorders[9,18] and dementia[9], in health psychology studies[19,20], and to address personality disorders[21]; however, these conditions were considered out of the scope of this Review. Similar immersive reality technologies, including augmented reality (referred to broadly as 'extended reality' or 'XR'), have some emerging potential in mental health[22]. However, the level of immersion of these technologies has not been researched extensively and they are at a relatively earlier phase of development than VR. Thus, our Review focuses only on highly immersive VR.

Active elements of immersive environments

Central to the effectiveness of VR-based therapies is the concept of immersion, which refers to the extent to which users feel mentally and emotionally absorbed within virtual environments[23,24]. The key factors influencing immersion include an immersive system, a sense of presence and ecological validity[3] (Fig. 2). By immersive system, we refer to the technical aspects of the VR equipment and software. Some examples of these technical aspects are visual fidelity (the degree of accuracy and realism in the visual representation), stereoscopic rendering (which creates a 3D effect in images by showing slightly different perspectives to each eye), spatial audio (which creates a sense of sound coming from different directions and distances to mimic real life), interactive feedback (the ability to respond to user actions in real-time) and user interaction modalities (the different ways in which users can engage with virtual environments).

By contrast, a sense of presence is the individual's perception of being in a given VR environment (place illusion) and experiencing the events that occur within that environment as believable (plausibility)[3]. Thus, immersion is determined by the objective capabilities of the system (immersive system) and the subjective response of the individual (sense of presence)[25]. Both aspects of immersion correlate highly with the degree to which VR accurately reflects behaviours and phenomena in natural, everyday settings (ecological validity)[26,27] and with the emotional[28], behavioural[23] and physiological response to the VR environment[29]. Traditional laboratory experiments often lack ecological validity because they isolate behaviours from their typical context, leading to findings that might not translate well to real-world situations. Ecological validity underscores the importance of designing studies that closely resemble the complexities of natural environments to ensure the relevance and applicability of research.

The setting in which VR is used also influences presence and immersion. For example, immersion might increase when a person uses VR in a space that they perceive as safe and fully private, such as a therapist's consultation room, compared to a space that is perceived as unsafe or public. Motivation and willingness to participate in the experience might also increase immersion. In this section, we expand on motivation and sense of presence as key elements of effective VR-based treatment.

Motivation

Although motivation is not specific to VR, it does have a central role in how individuals engage with VR environments. Motivation is a fundamental driving force behind human behaviour, influencing performance, productivity and goal attainment in various domains such as education, work, sports, and personal development[30]. Understanding and harnessing motivation can lead to improved outcomes, increased engagement and overall satisfaction in VR-based treatment.

Deficits in motivation are common across different mental health conditions and are closely linked to poor daily functioning[31].

Fig. 1 | Typical setup of therapist-delivered VR-based treatment. The client is immersed in a virtual reality (VR) environment by wearing a VR headset. The therapist launches and controls a VR program on a computer that sends updated information about the VR environment to the headset via Wi-Fi or cable. The therapist can view and change what the client sees and communicate with them via headphones and a microphone.

Consequently, adequately assessing and targeting motivation is a crucial part of the recovery process[31,32]. A common claim of VR-based methods is that this technology is engaging. Indeed, engagement can be achieved via gamification, immersion, and the structure and control that VR environments can provide[33]. This rationale has prompted the use of VR in the treatment of mental health conditions in which motivational deficits are a clinical symptom[34].

However, VR-based interventions do not specifically address motivational deficits and might influence a broader span of deficits associated with mental health conditions, including deficits in the capacity to experience pleasure (pleasure experience)[35,36]. A VR-based version of behavioural activation was found to be equivalent to traditional behavioural activation in treating major depression but superior to standard care[37]. This finding indicates that VR can effectively incorporate motivational aspects stemming from activation and exposure to everyday life occurrences, even in the absence of these experiences in real life. A VR-based therapy called V-NeST distinctly targeted motivational and pleasure experience deficits associated with the negative symptoms of psychosis[34]. Using VR environments with different motivational needs, this therapy provided opportunities for experiential learning on the use of positive feedback to promote goal-directed behaviour.

Motivational deficits also have a central role in theories of recovery from addiction. In this area, VR-based research has predominantly focussed on attentional bias and physiological reaction to addiction cues[38] and has shown that more realistic exposures and scenarios are associated with higher craving for alcohol and cigarettes[39]. Most addiction intervention studies using VR focus on cue exposure therapy, in which craving is targeted via exposure to visual cues of the desired substance. This paradigm targets motivation indirectly, similarly to serious games programs that tend to calibrate rewards and use pleasure experience to maintain therapy engagement[40].

In summary, motivational deficits and alterations are relevant to many mental health conditions; however, these deficits and alterations are addressed only indirectly by VR paradigms through engaging procedures or embedded enjoyment and gamification. Addressing motivational deficits in VR procedures will require systematic consideration of patient efforts in task completion and the extent to which reward and feedback drive behaviour.

Sense of presence and therapeutic change

People report experiencing some degree of presence in nearly all computerized environments, regardless of the level of immersion[41]. A higher sense of presence is linked to heightened user effects, such as the similarity between the responses to virtual stimuli and interactions in the real world[23]. Consequently, a higher sense of presence has been associated with more immersive VR applications[42,43].

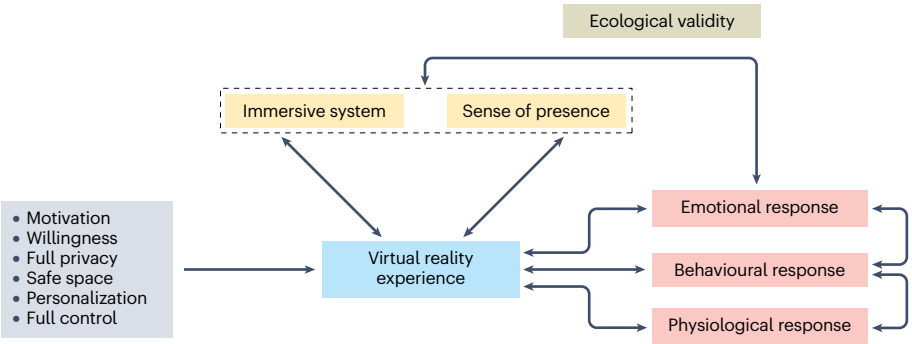

Fig. 2 | Active elements of immersion. The level of immersion of the virtual reality (VR) experience of an individual is determined by the objective characteristics of the VR environment (immersive system) and the subjective characteristics of the response of the individual to the VR environment (sense of presence). These elements improve the ecological validity of VR-based interventions, leading to specific emotional, behavioural and physiological responses. Motivation, willingness to use a VR environment, privacy and safety associated to the VR setting, personalization of the experience, and perception of control also increase immersion.

Although sense of presence is most frequently assessed using self-reporting questionnaires[44], there is a growing interest to measure physiological correlates of sense of presence in VR using techniques such as electroencephalography and functional magnetic resonance imaging[44]. Additionally, researchers are exploring techniques that enhance immersion in VR systems through adjustments known as configuration transitions. These adjustments involve altering aspects such as the layout, visual effects, audio cues or interactive elements within the VR experience to enhance immersion or improve user engagement[42]. Certain immersive technology components have been found to enhance a sense of presence, particularly tracking level (the degree of accuracy and precision with which a VR system can track the movements and positions of the head and hands of a user), stereoscopy (creating a 3D perception), field of view (the extent of the visible environment or scene that a user can see while wearing a VR headset) and multi-sensory feedback[43,45].

A potentially threatening real-life environment can be replicated in VR, which creates a safe setting where people can explore and act without feeling threatened[46]. Nothing the individual fears can really happen to them in VR environments. Such safety assurance, combined with a high sense of presence, enables individuals to freely explore, experiment, and experience feelings and thoughts[47]. This setting is particularly relevant for treatment because therapists can create controlled environments to gradually expose patients to their fears. This approach, often termed 'exposure therapy', enables patients to confront and work through their fears and anxieties in a secure and supportive context, which helps them to build resilience and overcome challenges more effectively. The extent to which a virtual environment can meet the expectations of users and trigger reactions depends on prior knowledge by researchers about the critical elements that a scenario must support to be relevant to the targeted individuals and the capacity to programme these elements.

Research has contributed to a better understanding of how a sense of presence can determine the outcome of VR-based therapeutic applications[45]. Place illusion and plausibility have a particularly crucial role in sense of presence[24]. The illusion of body ownership is another system-dependant illusion that contributes to a sense of presence and might be relevant for clinical applications. This illusion, which happens when participants see their virtual body substituting their own when looking down towards themselves, can help individuals with perceptual body distortions (for example, eating disorders)[24,48].

Some studies have shown the bidirectional relationship between a sense of presence and emotions. On the one hand, the higher the sense of presence, the higher the intensity of emotions experienced during the therapy. On the other hand, the higher the intensity of the emotions, the higher the sense of presence and reality judgement[49]. Using VR environments with emotionally relevant content to target specific emotions and related cognitions can increase clinical improvement[50]. Thus, feeling present during exposure might be necessary but not sufficient to achieve benefit from VR-based therapy. The interaction between a sense of presence and relevant therapeutic processes must be considered to understand clinical and meaningful change.

Treatment through VR

VR has been used to treat various mental health conditions. In particular, VR immersive and controlled environments have led to a growing body of research on VR-based exposure therapies and treatments involving skill building and symptom management (Fig. 3). Randomized controlled trials have demonstrated that these treatments can be effective[15]. Although the mechanisms driving these effects are still being studied, findings suggest that the effectiveness of the interventions might be due to engagement with the virtual environments and a sense of presence, which enable repeated and safe exposure and adaptation to relevant feared stimuli. In this section, we review evidence of well-established VR-based exposure therapies and other VR-based treatments for anxiety, psychosis, post-traumatic stress, eating disorders, depression and stress symptoms.

Anxiety

The use of VR to treat anxiety disorders is one of the largest and most developed areas of research[51]. Anxiety disorders, characterized by pathological fear and anxiety, include generalized anxiety disorder, social anxiety disorder, specific phobias and obsessive–compulsive disorder. These disorders account for 14.6% of disability-adjusted life years globally, indicating that they substantially reduce population health[52].

Findings consistently show that VR-based exposure treatments can reduce symptoms of anxiety disorders in adults[4] and that treatment effects generalize to real life[53]. Fear extinction is a key process in exposure therapy. Exposure treatment involves repeated confrontation with feared stimuli, either external (objects, people, situations) or internal (bodily sensations, memories, hallucinations), in the absence of the feared outcome. Fear extinction occurs when the feared stimulus (conditional stimulus) that was previously paired with an aversive outcome (unconditional stimulus) is presented without being followed by the aversive outcome. Experiencing the stimuli without that aversive outcome creates a corrective learning experience in which believability of the feared outcome and anxiety decrease (Fig. 4a). Through exposure

therapy, individuals learn that the feared object or situation is not as threatening as they initially believed, leading to a reduction in their anxiety response over time.

VR-based therapy for anxiety disorders has been shown to be more effective than treatment as usual or waiting-list control[51,54]. VR-based exposure therapy is non-inferior to conventional treatments (in vivo exposure) for most anxiety disorders[51]. Dropout rates do not differ between VR-based and in vivo exposure treatments for anxiety disorders[51], indicating that clients might accept both types of treatments equally. However, standard in vivo procedures might be more effective than VR-based exposure for some specific phobias (such as arachnophobia, panic disorder and agoraphobia), as VR might not yet reach the high level of immersion and presence of in vivo procedures for these exposures[55].

An approach that could potentially close the gap between VR-based and in vivo exposures is going beyond realism and using VR to create fictional situations, for example, spiders the size of a dog[56]. VR can be leveraged to enhance the therapeutic effects of exposure. For instance, in the case of confronting a giant spider, VR can amplify the disparity between the expectations of an individual and the actual outcome[56].

A systematic review found only two studies of VR-based treatment of obsessive–compulsive disorder[9]. The first study included 3 participants in a 12-session cognitive-behavioural therapy programme that involved 8 VR-based sessions (consisting of touching filthy walls and toilet bowls) and found preliminary evidence for symptom improvement from baseline to post-intervention[57]. The second study followed one participant during 60 sessions of VR-based exposure therapy over 2 months and found a reduction in symptom severity from baseline to post-intervention[58].

In sum, overall evidence supports the effectiveness of VR-based therapy for anxiety disorders. However, evidence gaps remain, especially for the treatment of panic disorder, agoraphobia and social anxiety disorders[9]. Additionally, there is a considerable lack of controlled trials testing the potential of VR-based therapies for the treatment of children and adolescents with anxiety disorders[59] (Box 1). Evaluations of the cost-effectiveness (assessing whether the benefits of a treatment or intervention justify costs involved in developing and delivering the intervention) of VR-based treatment for anxiety are also lacking[6,51,60,61].

Psychosis

Most of the evidence for the use of VR in the treatment of symptoms of psychosis focuses on paranoia and auditory hallucinations[10,61–63]. Overall, findings suggest that VR-based treatment for psychosis can be effective, safe, and cost-effective[10,62] and that client attitudes towards VR are positive[10].

VR-based treatment of paranoia is supported by the same model as anxiety disorders. Treatments aim to challenge the underlying assumptions that maintain paranoid beliefs through exposure to environments and scenarios that offer corrective experiences[10] (Fig. 4b). The selection of relevant feared social situations in VR environments is informed by a personalized case formulation, which involves collaboratively gathering information about the background, experiences, symptoms and current circumstances of a client to create a comprehensive picture of their unique situation[64]. During the VR exposure, clients are guided to refrain from using safety behaviours (such as avoiding eye contact or acting hostile) and to reshape their beliefs about safety and harm through positive learning experiences. Unfounded dysfunctional beliefs lose credibility across sessions as the client learns that they are safe, resulting in reduced anxiety and improved confidence in social situations. Several randomized controlled trials have shown that VR-based treatment reduces paranoid thoughts and anxiety[65,66] and that effects generalize to real life[66]. Evidence supports the efficacy of VR-based treatments for paranoia[67], but direct comparison to psychotherapy with in vivo exposure and long-term effects (more than 3 months follow-up) are lacking.

VR is also being used to treat auditory hallucinations. The application of VR and associated technology offers the opportunity to connect interventions with the lived experience of voice-hearing in a novel and potentially powerful way[68]. One of the most established approaches is AVATAR therapy, in which the client creates a digital avatar to embody their main distressing voice, personalized for visual and sound characteristics[69]. Voice-transformation software enables face-to-face dialogues between the client and the avatar to reduce

Fig. 3 | Example of VR-based exposure therapy. The virtual reality (VR) social environment in the treatment of paranoid beliefs and social anxiety exposes the client to illness-related challenges. For example in the Orygen Digital Revive study, one of the VR environments was a restaurant in which the client was guided to buy coffee (social interactions), was exposed to perceptual triggers (crowdedness) and coped with emotional content elicited by another customer (negative emotions).

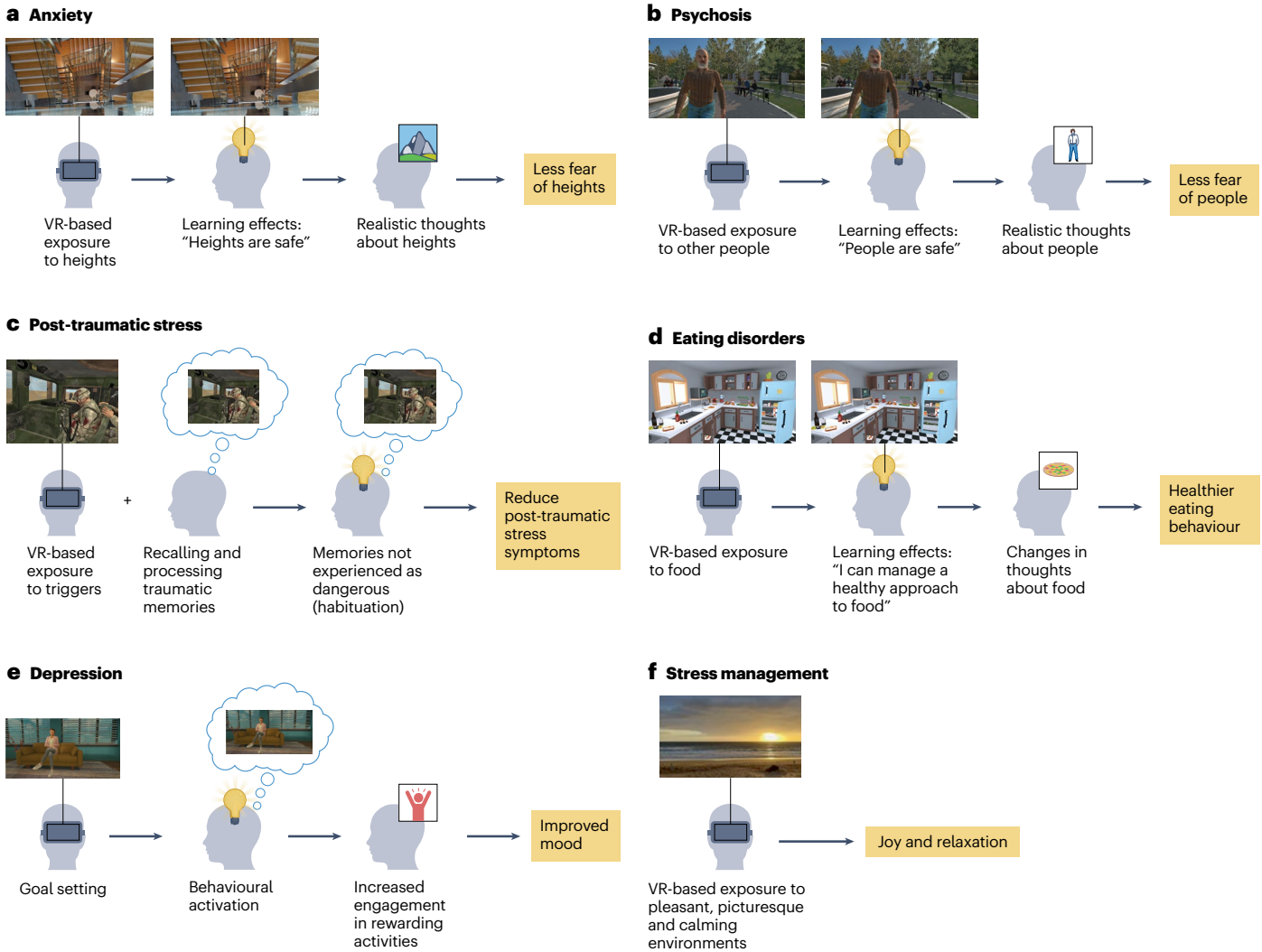

Fig. 4 | Examples of VR-based treatments. a, Virtual reality (VR)-based exposure therapy for anxiety is personalized to the client (for example, fear of heights), who is guided through a learning process to reduce the severity of anxiety symptoms. **b**, VR-based therapy for paranoid ideation focuses on the client's fears to gradually reduce paranoid thoughts. **c**, VR-based therapy for post-traumatic stress involves exposure to a VR environment resembling the traumatic event while the client recalls and processes the traumatic memories. **d**, VR-based therapy for eating disorders exposes clients to interact with food in a virtual kitchen towards developing healthier eating attitudes and behaviours. **e**, VR-based behavioural activation in depression gradually increases engagement of clients in rewarding activities to improve their mood and regain a sense of control and fulfilment in their lives. **f**, VR-based stress management provides immersive tranquillity for users by featuring serene natural environments such as beaches, forests and mountains.

anxiety and increase power, control, and confidence in dealing with the voice. This 2D virtual environment approach[70] led to reductions in voice frequency and distress compared to the post-intervention performance of an active control group[69,71]. The interaction between reduced anxiety and a strong sense of voice presence has been associated with positive therapy outcomes[72].

The implementation of standard non-immersive AVATAR therapy is continuing apace[73], including remote delivery through video conferencing platforms. However, the broader question of whether the use of full VR immersion confers specific benefits remains open. The first test of the AVATAR therapy approach in highly immersive 3D VR environments suggested a decrease in auditory verbal hallucinations and persecutory beliefs and improved quality of life. However, the tests of 3D VR environments of the AVATAR therapy involved small-scale studies[74,75]. A fully powered replication of the original randomized trial (AVATAR1)[71] using a highly immersive approach has been developed using separate software and is being tested in an independent trial[76].

Post-traumatic stress

A growing evidence base supports the use of VR for PTSD. In VR-based exposure therapy, the client narrates the traumatic event while being exposed to a VR environment that represents it. This approach is informed by evidence-based imaginal prolonged exposure strategies[77]. During imaginal exposure sessions, individuals are encouraged to

recount the traumatic event, including the thoughts, emotions and sensory experiences associated with it, as if they were reliving the event. The goal is to desensitize the individual to the traumatic memories and reduce the distress and avoidance behaviours associated with them. Prolonged exposure refers to the gradual and repeated exposure to the traumatic memories over time, which enable the individual to habituate to the distressing memories and learn that they are no longer dangerous. VR-based exposure therapy provides a low-threat context in which the client can confront and therapeutically process the emotions related to the traumatic event and de-condition the stress response via a fear extinction learning process[78] (Fig. 4c).

Most clinical trials have tested the efficacy of VR-based exposure therapy for the treatment of combat-related PTSD[79–94]. The consensus from this research suggests that VR-based exposure therapy is associated with better outcomes compared to treatment as usual[88,95], and it is as effective as other proven treatments, including imaginal prolonged exposure[86,92]. Two randomized controlled trials demonstrated that VR-based exposure therapy and imaginal prolonged exposure led to equivalent positive post-treatment outcomes (assessed with the gold standard outcome measure the Clinician-Administered PTSD Scale)[85,96]. However, in one of the trials, imaginal prolonged exposure produced greater symptom reductions at 3-month and 6-month follow-up compared to VR-based exposure[96].

The difference in symptom reduction might be in part due to the limited variety of VR scenarios available at the time of the trial. Indeed, the VR environment showed only four different virtual scenarios, all from Iraq, which might have limited the customization of content needed to address diverse client experiences (such as deployments in Afghanistan or other contexts). This limitation was remedied in the second trial[85], in which system content offered 14 Iraq and/or Afghanistan virtual contexts. The results of the second trial showed that positive post-treatment outcomes were equivalent across therapies at all follow-up time points. Moreover, VR-based exposure therapy outperformed imaginal prolonged exposure in patients with PTSD and comorbid major depression across all follow-up time points. Finally, 76.7% of research participants said that they would have chosen VR-based exposure therapy over imaginal prolonged exposure therapy if they were to have a choice of treatment (all participants knew they would be randomly assigned to a treatment condition).

The findings from these trials are instructive on three points. First, customization of VR scenarios that are more relevant to the trauma experience of an individual might lead to better outcomes. Second, individuals with comorbid PTSD and major depression, for whom employing imagination to confront their traumatic memories might be challenging, could find value in the emotionally immersive nature of VR environments. Third, VR-based exposure therapy might be more appealing to clients than traditional treatment formats, which can help to break down barriers to care.

Finally, multiple studies have reported positive outcomes using VR-based exposure therapy for PTSD not related to combat. For example, survivors of the attacks on the World Trade Center towers in the USA[82] or sexual trauma survivors[86] with symptoms that, in many cases, were unresponsive to a previous course of imaginal prolonged exposure treatment, benefitted from VR-based exposure therapy. Insights from these findings are relevant for the design of future studies and for informing treatment recommendations based on the clinical characteristics and treatment preferences of a client. For example, multi-modal, motion-assisted, memory desensitization and reconsolidation therapy is a therapeutic approach that integrates elements of eye movement desensitization and reprocessing therapy with VR technology. Although the evidence for this approach is still emerging[95], the aim is to enhance the effectiveness of trauma treatment and provide an engaging therapeutic experience for clients. Together, the overall trend of results encourages the clinical use of VR-based exposure therapy as a safe and efficacious evidence-based treatment for PTSD.

Eating disorders

Eating disorders are mental health conditions characterized by abnormal habits, attitudes and behaviours related to eating, food and body image. Patients with eating disorders frequently show high ambivalence towards changing their attitudes and eating habits. Unfortunately, the recovery rates for eating disorders are only between 40% and 60%[97]. Learning new eating habits, attitudes, and behaviours along with

Box 1 | VR-based treatment for adolescents

Cross-sectional analyses underscore a concerning prevalence of mental disorders among children and youths aged 5 to 24 years worldwide, with over 1 in 10 (293 million) young individuals affected[167]. These disorders account for approximately one-fifth of all disease-related disability in this age group, and this demographic represents roughly one-fourth of the total mental disorder burden across all age groups. Despite the growing need for evidence-based interventions in this group, adolescents face substantial barriers to accessing services. Approximately 70–80% do not seek professional care[168] and only 1 in 10 adolescents among those who seek help receive appropriate care[169]. Moreover, treatment engagement is challenging in this age group, with treatment adherence as low as 40%[170].

Virtual reality (VR) presents promising opportunities for enhancing experiences and interactions, particularly in therapeutic contexts, providing an immersive and engaging environment that can captivate adolescents. These characteristics make therapy sessions more appealing and can potentially increase the participation and motivation of adolescents in treatment[171]. Nevertheless, there are notable research gaps within adolescent populations[16,59]. Studies investigating the effects of VR on social skills, empathy and emotional development in children and teenagers are limited, compounded by the absence of standardized treatment protocols[172]. Additionally, although adolescents generally enjoy and engage with VR-based games, they might not consistently perceive VR as directly applicable to addressing their mental health issues[16].

Ensuring the validity of interventions conducted in VR environments and comprehending the effect of VR on cognitive functions, such as attention, memory and learning, in young individuals require further exploration[172,173]. VR is displayed using two screens close to the eyes, and concerns have been raised about the effects of prolonged exposure to VR on the developing visual system and eye problems in the short term[174]. Technological advances have resulted in displays that provide a captivating 3D experience that the visual system processes similarly to real-life experiences[175]. Moreover, VR has been used to correct visual impairment (amblyopia) in children[176].

repeated practice are needed to consolidate therapy-driven changes and to enhance the effectiveness of treatments.

VR is one of the earliest forms of digital technology tested in the field of eating disorders[98–100]. Studies have employed this technology to measure or change patient behaviours and attitudes towards illness-related cues such as food, body image and body shape[98,99,101]. Exposure to virtual food elicits similar emotional reactions to exposure to real food, and higher emotional reactions than food pictures[102]. Similarly, exposure to virtual food elicits higher levels of negative emotions and disorder-related attitudes and behaviours (such as anxiety, body dissatisfaction and craving) in patients with eating disorders compared to control individuals[98,99].

Therapists can guide patients through virtual scenarios that challenge negative thoughts and beliefs about body image and eating habits, which promotes healthier attitudes towards food (Fig. 4d). VR creates controlled environments where patients with eating disorders can confront realistic body images to address dissatisfaction with their physical appearance. Randomized controlled trials have been conducted to test the use of VR to deliver, supplement or enhance standard psychological treatments for eating disorders[9,98–100]. In patients with binge-type eating disorders, VR-based therapy was associated with a reduction of core eating disorder symptoms such as episodes of loss of control over eating behaviours. However, studies have yielded inconclusive findings regarding the accuracy of VR in assessing body image and body dissatisfaction[100].

Preliminary findings about the use of VR-based treatments for anorexia nervosa are also promising. VR-based exposure therapy using virtual food was associated with reduced anxiety surrounding eating, enhanced comfort around food and a decrease in food-related fears[103–105].

Finally, a fast-growing field of application is the use of VR to treat perceptual body distortions and body representations through the embodiment of avatars of different sizes and shapes and the induction of body size illusions[99]. The term 'bodily illusions' refers to phenomena in which the brain integrates information from multiple senses to create a perception of the body that might not accurately reflect reality[106]. By combining insights from bodily illusions and leveraging VR technology, applications have emerged for the evaluation and treatment of body distortions in eating disorders[99]. VR environments can simulate different body shapes and sizes, enabling individuals to experience their bodies in diverse ways. Therapists can use these VR experiences as part of treatment interventions to help patients to challenge and correct their distorted body image perceptions[107]. This type of approach is promising and subject to further investigation[99].

Depression

Depression is a mental health condition characterized by persistent feelings of sadness, hopelessness, and a loss of interest or pleasure in activities once enjoyed. It can also manifest through physical symptoms such as changes in appetite or sleep patterns, fatigue, and difficulty concentrating. Depression can substantially impair daily functioning and often requires professional treatment, including therapy and medication[108].

Drawing meaningful conclusions about depression-related outcomes is challenging due to the limited number of studies in this area[109]. Given the intricate and multifaceted nature of depression aetiology and maintenance, developing VR-based interventions that comprehensively address all facets of depression might not be feasible. A more promising approach could entail utilizing VR-based interventions to target symptoms of depression within specific population cohorts. For instance, a VR-based therapy for depression in clients who experienced childhood adversity and trauma would focus on fostering self-compassion and reducing critical self-judgement[110], and another VR-based therapy for depression to help clients to learn new coping strategies would focus on enhancing awareness of automatic thought patterns by observing examples from others[111]. In patients with stroke, VR-based rehabilitation exercise and rehabilitation of neurological function led to a substantial decrease in symptoms of depression[112]. Behavioural activation focuses on helping individuals engage in meaningful and rewarding activities to improve their mood and overall well-being. By increasing activity levels and reconnecting with enjoyable activities, individuals can experience a reduction in depressive symptoms and an improvement in their quality of life[113]. VR-based behavioural activation was found to be feasible and well received among patients diagnosed with major depressive disorders[37] (Fig. 4e).

Stress management

VR-based relaxation is an innovative and accessible low-intensity psychological intervention associated with increased psychological well-being, stress management and mental health protection in the general population[114] (Fig. 4f). The virtual environments used in VR-based relaxation studies are typically pleasant, picturesque and calming environments, often depicting graphics or videos of the natural world such as beaches, oceans, forests, meadows, mountains, lakes and waterfalls. Some environments include interactive, gamified components, such as mindfulness or guided meditations, or games and activities that enable individuals to playfully interact with features of the environment. Intervention time is typically brief and often offered as a single session. Interventions can be facilitated by a therapist or administered by the user as a self-help intervention. Self-help applications of VR-based relaxation for the home or in the workplace[17] have focused on people in high-pressure jobs or key workers, such as healthcare professionals, including mental health staff, who can experience burnout and compassion fatigue[115,116].

VR-based interventions for relaxation are feasible, acceptable and potentially effective short-term interventions to promote relaxation and reduce stress in people with mental health conditions[17]. Research indicates that exposure to real-world nature has many physical and mental health benefits[117]. However, natural environments might be inaccessible to patients with severe mental health conditions due to a range of factors, for example, financial restrictions, severe anxiety or paranoia in open or social environments, physical disabilities or chronic conditions, living in cities where there is a lack of green natural spaces, or because they are staying in restrictive environments such as psychiatric hospitals. Delivery of VR-based relaxation focused on simulated picturesque natural environments might be particularly powerful and useful for these patients[118].

The largest studies have focused on outpatients and coupled VR-based relaxation with other interventions such as meditation for outpatients with PTSD ($n = 96$)[119] or physical activity for outpatients with anxiety ($n = 60$)[120]. Small studies have also been conducted with patients with conditions such as eating disorders ($n = 20$)[121], depression ($n = 17$)[122], bipolar disorder ($n = 20$)[123] and psychosis ($n = 25$)[121–124]. There is some initial evidence that VR-based relaxation might reduce levels of violence and aggression among inpatients on psychiatric wards and reduce the number of restrictive practices administered by ward staff[118]. One pilot study found that playing a VR-based relaxation app was associated with a calm or relaxed state that lasted 2 h on average in adolescent outpatients at a specialist secondary school with psychiatric

and behavioural problems (including autism and attention deficit hyperactivity disorder)[125].

Key strengths of VR-based relaxation are its accessible self-help and transdiagnostic applications for stress and anxiety management. Limitations of the current research include limited standardization across studies of fundamental aspects of the interventions, such as intervention type, VR environments, duration of the VR sessions, number of sessions and outcome measures. There is also a lack of large randomized controlled trials, studies on the longer-term effect of VR relaxation, and causal data linking real-world behaviour change and VR-based relaxation. Nevertheless, there is enormous potential for VR relaxation as a low-intensity stress management intervention in mental health clinics or psychiatric wards, or as a self-help home-based intervention.

In summary, there is growing evidence for the efficacy of VR-based treatments for anxiety disorders, PTSD and psychosis. Some evidence is emerging for the use of VR-based treatment in eating disorders, depression and stress reduction; however, larger and controlled trials are needed before drawing definitive conclusions.

Emerging areas

VR research has historically focused on exposure-based treatments delivered under the supervision of a therapist using virtual environments that resemble the real world. Technological advances have brought new opportunities to extend and enhance VR capabilities to deliver more tailored, targeted and engaging treatments. Within the VR headset, a comprehensive network of sensors, including heart-rate monitors, eye-tracking cameras, head-movement sensors (accelerometers, gyroscopes, magnetometers), hand-tracking technology, and facial expression recognition cameras, continually collect a wealth of physiological and emotional data from the individual. These diverse data streams can then be analysed to interpret responses to the VR environment and adapt the treatment accordingly (Fig. 5). For example, eye-tracking studies have classified user emotions with 60–90% accuracy[126] and heart-rate monitors, such as smart bracelets, can be used to characterize clinically relevant mood states[127]. In this section, we discuss the integration of multiple sources of data using artificial intelligence and other emerging areas for VR-based therapy, including biofeedback and therapeutic VR games.

Artificial intelligence

Artificial intelligence is a broad term that captures a range of techniques, including machine learning and natural language processing. These techniques harness algorithms and computer systems to analyse large amounts of data to detect patterns and make predictions. The level of therapy personalization is a known driver of treatment

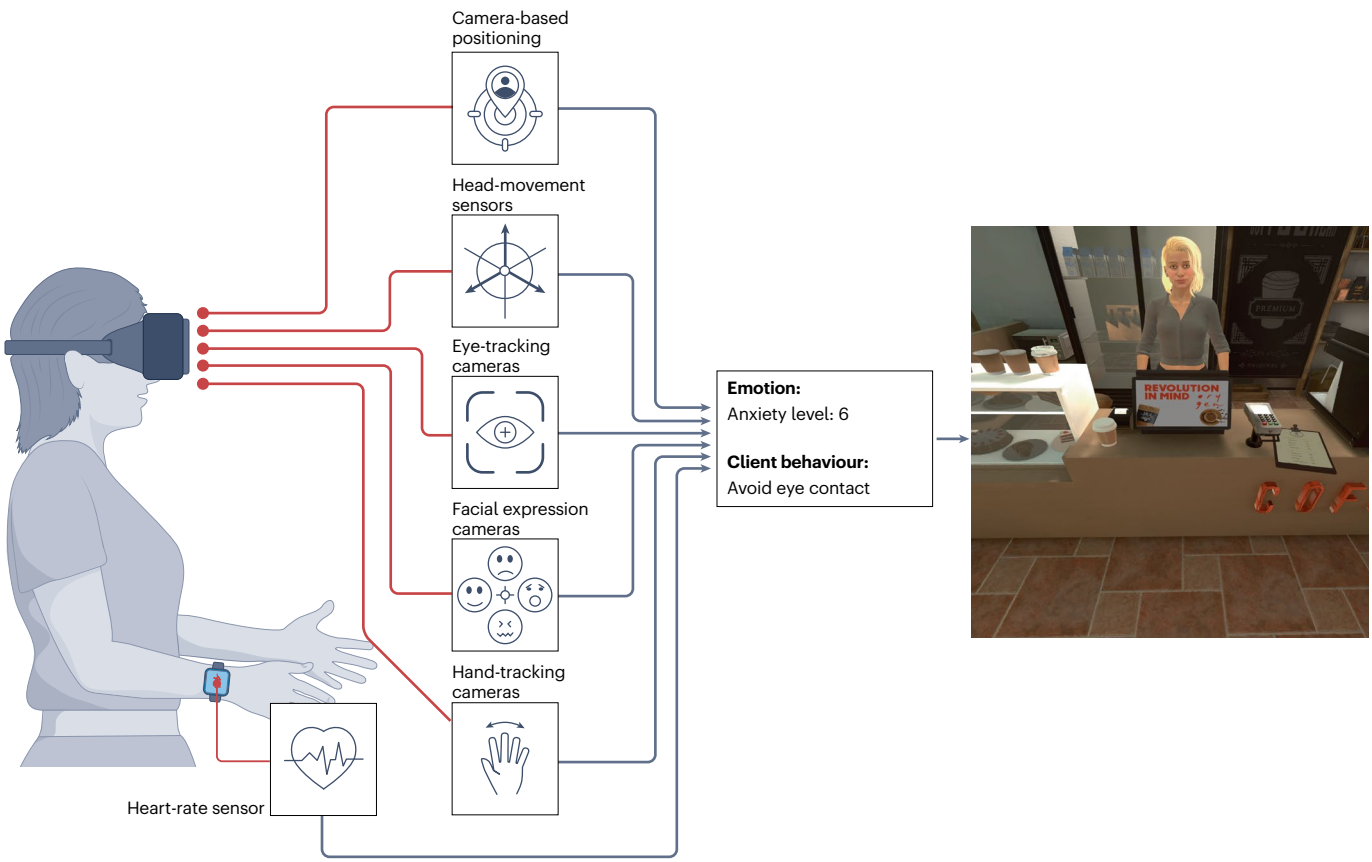

Fig. 5 | Sensors and cameras used for real-time updates of the virtual environment. A comprehensive network of sensors can be included in the virtual reality (VR) headset that helps to track client responses to the VR environment. Information on the person's position, their head, eye and hand movements, facial expressions, and heart rate can be collected through these devices. These data sources are then analysed to interpret client responses to the VR environment and adapt the treatment accordingly.

effects[128], and adaptation through data tracking will be an important future direction in VR-based therapies. For example, integrating facial recognition software[129] into psychological treatments might be an innovative way to improve outcomes. Through integrated systems that provide real-time analysis of data in the VR headset, facial expressions and other physiological patterns indicative of stress, anxiety, or discomfort can be rapidly detected, informing real-time decisions to provide a dynamic and highly personalized therapeutic journey. In a VR setting, the therapist can update the environment (for instance, by adding more people and background sounds) to modify the level of anxiety of a client based on data about their emotional state.

An area of particular potential is the use of artificial intelligence to automate the delivery of VR-based treatments, which has important implications for increasing the availability of effective treatments and accessibility in often overburdened mental health-care systems. In automated VR-based treatments, individuals are typically guided by an automated avatar and the computer system has some capability to personalize and direct the treatment based on their response to different scenarios. However, automated personalization is currently limited to basic decision trees that are restricted in the degree to which they can adapt the treatment over time[6]. Developments in artificial intelligence have the potential to overcome this limitation by harnessing real-time data within the headset to track and adapt the therapy automatically, much like a clinician would in everyday practice. However, clinician attitudes around trust, confidence and system transparency might be a potential barrier to widespread adoption[130].

Beyond personalization, automated VR-based therapy enables self-guided treatment that can be accessed independently from a clinician, which might increase the scalability of mental health solutions. More research is needed to establish the safety and efficacy of this approach, including for which populations it might be suited. Basic forms of artificial intelligence have already been integrated into VR-based treatments for fear of heights[131,132], public speaking anxiety[133], spider phobia[134] and psychosis[135,136], with good research evidence for effectiveness. For example, an automated VR-based therapy targeting fear in social situations (gameChange VR) was compared to treatment as usual in a randomized controlled trial including 174 people diagnosed with a psychotic condition and experiencing agoraphobia[135]. An automated avatar (virtual coach) guided the progress in the active arm through increasingly challenging situations by encouraging participants to refrain from using safety behaviours and providing feedback. Although there was a mental health-care worker present during intervention delivery, they did not have training in cognitive therapy and had minimal therapeutic involvement. Participants in the active arm showed a greater reduction in avoidance behaviours compared to the control group and this was mediated by improved safety beliefs and reduced defence behaviours. When delivered at scale, automated VR-based treatments, such as gameChange VR, could improve access to effective treatment, potentially at a reduced cost compared to treatment as usual[137].

Biofeedback within virtual environments

Biofeedback is a technique that involves monitoring physiological signals (such as heart rate, muscle tension, skin temperature or brainwave activity) collected through sensors placed on the body (such as the skin, scalp or fingers), and providing real-time feedback to individuals about these bodily processes. Through this feedback, individuals can learn to control or manipulate their physiological responses to improve health and well-being[138]. For example, patients can learn to identify and alleviate anxiety by receiving immediate feedback from biosensors that monitor muscle tightness or heart rate[139].

Attention is crucial in biofeedback training because it enables individuals to consciously observe and interpret feedback. By paying attention to how their physiological signals change in response to certain thoughts, behaviours or relaxation techniques, individuals can learn to control or manipulate these signals to achieve the desired outcome[140]. Difficulties with engagement and maintained attention are common challenges to biofeedback effectiveness[141]. VR offers a potential solution to these challenges by providing appealing and immersive content, and inducing anxiety levels comparable to real-life situations.

Incorporating gamification elements in VR environments can enhance motivation, and immersive hardware like VR glasses minimizes environmental distractions, which promotes attentional focus on the therapeutic task[141]. For example, the DEEP application immerses users in an ocean environment where they can control their movement by regulating their breathing, which is measured continuously using a chest band. University students with elevated anxiety showed improvements in anxiety symptoms following four sessions using the DEEP application and showed greater levels of engagement and self-efficacy relative to a control condition[142].

Therapeutic virtual games

Using gamification in VR environments to boost engagement and enjoyment of therapy has also been relatively unexplored. In the broader digital mental health literature, gamification elements, such as rewards, feedback or challenges, have been used within smartphone apps, online therapy programmes, and wearables to motivate and engage users[143,144]. For example, a non-immersive computerized VR game called SPARX used storytelling to help young clients to develop and practice therapeutic skills through exploration, puzzles, games and quizzes[145]. A non-inferiority trial found SPARX was equally effective as usual treatment in reducing depressive symptoms amongst adolescents, with good retention rates, and patients reported that the games helped with their engagement[146].

VR-based therapeutic games are increasingly prevalent both outside and within research environments. By 2020, there were 182 games with some link to mental health in the STEAM store (a large VR app store)[147], most of which were focused on mindfulness and meditation. However, none of the games had clear links to research studies, and some contained concerning content such as aiming to induce digital hallucinogenic experiences. This observation highlights the need for more rigorous research and oversight. With the proliferation of digital health products available directly to consumers, many of which make strong claims without sufficient scientific evidence[148], there is a need to address growing concerns about ethics and regulation in the digital mental health field. Issues pertaining to privacy, data security and safety are well documented in the broader digital mental health literature[149–152]. These issues are equally relevant to VR, where the collection and storage of potentially sensitive digital information raises important security questions.

Summary and future directions

Although research into VR was first established in the late 1960s, the field is still developing as technological advances rapidly evolve (Box 2). There is good evidence for the feasibility and acceptability of VR across several clinical populations. Similarly, good evidence exists for the effectiveness and efficacy of VR-based cognitive-behavioural treatments for anxiety disorders, psychosis, and trauma-related conditions

and evidence is emerging for eating disorders, depression and stress reduction. However, more high-quality randomized controlled trials are required to fully establish the efficacy of VR-based treatments across different populations.

Little is known about the adverse effects of VR-based therapies[153], although results from randomized controlled trials suggest that major adverse effects are no more likely than in routine treatment. Nonetheless, the harms or discomfort associated with VR-based therapies might be more nuanced than traditional safety monitoring protocols find acceptable. For example, VR experiences can be disorienting for some people, and the sensory mismatch between visual perception and physical motion when using VR technology might lead to some people experiencing a motion sickness characterized by nausea, dizziness and headache (cybersickness). Although improvements in hardware and software have decreased the chances of cybersickness, there is still a lot unknown about human factors such as age, gender and other demographic features that might influence cybersickness[154]. Similarly, some individuals report feeling disconnected from their thoughts, feelings, memories or sense of identity (dissociative experiences, often as a response to trauma or stress) when using VR, and it is currently unknown whether these experiences have long-term effects[155,156].

The ability to control exposure to feared stimuli in VR is key to its therapeutic potential. However, this level of control also enables potential harm if not carefully considered. Negative emotions resulting from VR have been positively correlated with negative rumination several hours after the VR-based intervention[157] (for example, self-criticism and shame related to a behavioural response in a VR environment). Thus, the therapist must help the client to manage and process emotional experiences emerging from the use of VR. There is a clear need for further research to more thoroughly understand the potential negative effects of VR exposures. Establishing appropriate clinical protocols with sufficient training and supervision will be important for protecting against any potential harm.

Digital interventions that perform well in clinical trials do not necessarily translate to strong uptake and engagement beyond the research context[158] as was already observed in the first generation of digital mental health technologies[158,159]. Researchers must work alongside the end users to confirm that these technologies are fit for improving outcomes for people with mental health conditions, and ensure that the system is adequately trained and supported to translate the great potential of this technology. Employing methodologies such as user-centred design in the development of new interventions is critical to ensuring that real-world needs and perspectives are accounted for in this process[159]. Implementation models such as the NASSS (non-adoption, abandonment, scale-up, spread, sustainability) framework[160] can be used to guide development and determine the likelihood that a digital solution will be successful in the real world.

The successful implementation of VR within clinical settings hinges on how VR fits within existing systems and workflows[160,161]. Most VR setups require a headset, computer and headphones, which is expensive. Setting up the equipment can also be time-consuming and requires some level of experience. However, VR apps are progressively becoming more user-friendly, and it is possible to create a mobile setup of VR equipment on a trolley or use relatively low-fidelity programmes within the headset that do not require a connection to a computer. Future clinics might have dedicated rooms to deliver VR-based treatments for optimum efficiency. Adequate training and resources are also needed to upskill clinicians to deliver VR-based treatments[162] and to educate and build awareness among potential users about evidence-based treatments. Although clinicians generally recognize the potential of VR to improve treatment outcomes, practical issues (such as cost, time and physical space), training, appropriate treatment protocols (including safety), and organizational structures are perceived barriers amongst service providers[163–166]. In addition, perhaps the biggest implementation barrier might be the limited availability of evidence-based VR therapy programmes. The number of available programmes is growing[147] but few have been scientifically validated, and the software is costly to build and maintain, with many companies yet to establish sustainable sources of funding.

Research priorities in the field of VR for mental health focus on expanding scientific understanding, improving interventions, and

Box 2 | Origins and development of VR-based treatments

Psychologists began exploring the clinical applications of virtual reality (VR) in the early 1990s. A pivotal moment occurred in 1995 upon the publication of a groundbreaking study using VR to treat acrophobia (fear of heights)[177]. Therapists mainly employed VR headsets to guide exposure-based interventions. Despite the limited visual and software quality, the results showed positive treatment outcomes.

Since the early 2000s, more complex and tailored VR-based interventions have emerged, and two primary methods are available to create immersive VR. The first method employs VR headsets, also known as head-mounted displays. These wearables, fitted with two screens (one for each eye), generate depth perception through slight projection variations. Many head-mounted displays incorporate movement sensors, enabling the virtual environment to respond to user actions and cultivating a sense of presence. The first VR headset dates back to the 1960s[178], and was so heavy it required ceiling suspension for use. Until 2013, VR headsets remained scarce and costly, and were almost exclusively found in university settings. A transformative moment occurred in 2013 with the release of the affordable Oculus Developer Kit 1. Since the development of this tool, high-quality, user-friendly consumer VR headsets have proliferated.

The second method to deliver immersive VR involves a 'cave automatic virtual environment' (CAVE), pioneered in 1992 (ref. 179). This environment is equipped with multiple high-resolution projectors that display stereoscopic images onto the enclosure walls, floor and, sometimes, ceiling. Various tracking systems are employed to track the position and movements of users within the environment. Tracking systems might include infrared cameras, motion sensors or electromagnetic trackers that monitor the position and orientation of specially designed tracking devices worn by users. This environment uses stereoscopic techniques to create depth perception in the virtual environment. Displaying slightly different images to each eye simulates how humans perceive depth in the real world, enhancing the sense of immersion.

Despite three decades of psychological research, clients have yet to fully benefit from VR-based tools in clinical settings. The future holds promise for overcoming this challenge as affordable VR hardware, advanced software and comprehensive clinical training become mainstream.

Box 3 | Priorities for future VR-based research

User experience and accessibility

- Embed perspectives of individuals with lived experience throughout all research phases, from conception to implementation, to guide the research process.
- Employ appropriate participatory design methods, such as user-centred design and codesign with end users (typically people with lived experience and clinicians), to ensure virtual reality (VR) treatments are fit-for-purpose and engaging with excellent user experience.
- Actively address equity and accessibility in VR treatment design, especially for diverse groups with specific accessibility requirements or from varied cultural and linguistic backgrounds.
- Ensure that VR technology is accessible and affordable across diverse socioeconomic backgrounds to minimize disparities in access to mental health care.
- Prioritize lived experience leadership within research teams to ensure research decision-making is responsive to the needs of end users.

Diversity and inclusion

- Involve a diverse range of individuals, including those from different cultural backgrounds, in all phases of the design, development and testing process to ensure cultural sensitivity and authenticity.
- Ensure the design of VR treatments is appropriate for diverse and representative lived experiences.
- Design VR experiences that can be customized to reflect diverse cultural backgrounds, enhancing equality and enabling users to feel represented and comfortable within the virtual environment.
- Include diverse characters and settings in VR environments that accurately represent various cultural backgrounds, identities and lived experiences.
- Engage with communities to gather feedback and insights on the relevance and effectiveness of VR-based interventions within different cultural contexts.

Standardized study reporting

- Develop and promote guidelines for standardized reporting in VR research, ensuring consistency, transparency and replicability (for example, the RATE-VR guidelines[180]).
- Transparently report and acknowledge lived experience involvement in research within publications, including relevant limitations.

Effectiveness and mechanisms

- Undertake randomized controlled trials to establish the evidence base for VR-based treatments across demographic groups and clinical contexts, particularly youth populations.
- Utilize innovative methods to deconstruct VR-based treatment mechanisms, including active control comparisons, mediation analyses, dismantling studies (systematic breakdown of a complex intervention into its individual components to compare their relative contributions to the overall outcome[181]) and real-time data capture techniques like ecological momentary assessment, eye tracking, and biosensors.
- Conduct trials to establish the cost-effectiveness of VR-based treatments compared to traditional methods.

Optimum personalization and tailoring

- Identify suitable candidates for VR-based treatments through subgroup analysis and individual response predictors, including qualitative insights from patients and clinicians.
- Investigate the effect of VR-based treatment personalization and the extent of tailoring required for optimal outcomes.
- Explore personalization and tailoring driven by artificial intelligence in VR-based interventions.

Novel treatments and assessments

- Research new intervention targets within VR such as representation and interaction with cognitive and emotional states.
- Explore the use of VR for novel therapeutic paradigms such as acceptance and commitment therapy and mindfulness.
- Integrate VR with cutting-edge technologies like artificial intelligence, gamification, biosensors and real-time data capture.
- Assess the feasibility and efficacy of self-administered VR-based treatments and the required level of clinician involvement.
- Conduct pilot tests within diverse communities to assess the cultural appropriateness and acceptance of VR-based mental health interventions.
- Develop and explore the impact of different VR delivery mediums, including smartphone apps.

Ethical and safety considerations

- Investigate and address ethical concerns, particularly in the context of the integration of artificial intelligence in VR environments.
- Study the safety profile of VR usage, focusing on adverse events and developing protocols for reporting these in research publications.

Clinical implementation and training

- Conduct implementation trials to facilitate clinician adoption of VR in mental health.
- Identify and address barriers to widespread clinical adoption, including technological, financial and training barriers.
- Explore strategies to increase patient engagement and retention in VR treatment.
- Develop resources and guidelines to enable clients to make informed decisions about VR-based treatment supported by clear and transparent expectations.
- Develop comprehensive training programmes and resources for clinicians to effectively implement VR.
- Engage policy advisors to produce guidelines on the implementation of VR.

Industry and clinical partnerships

- Strengthen collaboration between researchers and industry to ensure the scalability and sustainability of VR solutions.
- Collaborate with clinical service providers to develop models for integrating VR-based treatments within routine clinical settings.
- Develop funding models and engage with policy advisors to support long-term VR-based treatment implementation.

ensuring ethical and effective use (Box 3). Key priorities include exploring how VR experiences can be inclusive and representative of diverse cultural, social, and economic backgrounds and involving people with lived experience in the conception, development and testing of new VR-based interventions to create treatments that are more engaging, resonant, and relatable for users, potentially increasing their efficacy. Investigating the long-term effect of VR-based interventions on mental health outcomes and examining their sustainability over time is also crucial to successfully implementing this research in clinical settings. Regulatory standards and ethical guidelines should be reformulated. Moving forward, research should focus on developing, deploying and evaluating VR-based mental health interventions tailored to individual needs and specific conditions, ensuring personalized treatments.

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